

VOCAWATCH

Design Thinking Project 2026

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USER / PROTAGONIST

Primary Users

- Student singers
- Casual vocal performers
- Public speakers
- Content creators
- Individuals with frequent voice usage

User Characteristics

- Use voice continuously for singing, practice, or speaking
- Often unaware of early vocal strain symptoms
- Users continue voice usage despite early discomfort
- Depend on subjective feeling instead of measurable feedback

Common Behaviors

- Ignore dryness or hoarseness
- Delay rest and hydration
- Practice for long durations
- Lack structured vocal care habits

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CONTEXT & SITUATION

Vocal strain is a growing issue among singers and frequent voice users. Most people recognize vocal damage only after symptoms such as hoarseness, throat dryness, reduced vocal range, or discomfort become severe.

Current preventive practices are reactive rather than preventive. Users rely on personal judgment and temporary remedies like hydration or rest instead of objective monitoring. Our research identified a major gap:

There is no accessible wearable system that provides real-time feedback and early detection of vocal overuse.

VocaWatch aims to bridge this gap using a wearable choker-style monitoring device that tracks vocal activity, environmental conditions, and usage patterns to encourage healthier vocal habits.

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EVIDENCE BASE

Research Methods

- Primary interviews with college singers
- Secondary research on vocal strain and vocal cord damage
- Empathy mapping
- User journey analysis
- Pain point analysis
- Root-cause identification
- Solution requirement checklist

Key Findings

- Many users overlook early signs of vocal strain.
- Users often recognize vocal strain only after symptoms become severe.Overuse is often unintentional
- Accessible and continuous vocal strain monitoring tools remain limited for everyday users.
- Users rely on subjective feelings instead of measurable data

Real-World Evidence

Professional singers such as Adele have experienced vocal damage requiring extended recovery, highlighting risks of prolonged strain.

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PAIN POINTS & NEEDS

Pain Points

- Lack of early strain detection
- No real-time feedback system
- Ignoring early symptoms
- Reactive recovery methods
- Inconsistent vocal care habits
- Dependence on subjective feeling
- Environmental stress factors
- Unintentional voice overuse

User Needs

- Early warning system
- Real-time monitoring
- Simple and wearable solution
- Preventive guidance
- Quantifiable vocal usage data
- Easy interpretation for non-professional users

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ROOT CAUSES (5 WHYS)

Why does vocal damage occur?

Because users continue speaking or singing despite strain.

Why do they continue?

Because early symptoms are underestimated or ignored.

Why are symptoms ignored?

Because there is no measurable feedback system.

Why is there no feedback?

Existing solutions are reactive and not wearable or continuous.

Root Cause Identified

Lack of accessible real-time monitoring and early-stage vocal strain detection.

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HOW MIGHT WE...?

- How might we detect vocal strain before serious damage occurs?
- How might we provide real-time feedback without interrupting users?
- How might we encourage preventive vocal care behavior?
- How might we objectively monitor vocal usage patterns?
- How might we combine environmental and vocal data into one wearable system?
- How might we make vocal health monitoring accessible to casual users?

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TOP IDEAS

VocaWatch

YOUR VOICE. OUR PRIORITY.

TOP IDEAS

Proposed Solution: VocaWatch

A wearable choker-style smart device that monitors vocal usage and detects early strain symptoms.

Key Features

- Voice intensity tracking
- Usage duration monitoring
- Pitch/activity analysis
- Throat vibration sensing
- Environmental monitoring
- Humidity
- Air quality
- Risk classification system
- Vibration alerts
- LED feedback indicators
- Preventive recommendations

Intended Outcomes

- Reduce vocal overuse
- Improve awareness
- Encourage healthier vocal habits
- Prevent long-term vocal damage
- Improve awareness of unhealthy vocal habits through objective

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PROTOTYPE / FLOW DIAGRAM

System Workflow

1. User begins speaking/singing
2. Sensors collect vocal and environmental inputs
3. System analyzes intensity, duration, and patterns
4. Strain level is estimated
5. Risk classification generated:
 - Safe
 - Warning
 - Danger
6. User receives feedback through:
 - Vibration alerts
 - LED indicators
 - Actionable suggestions
7. User adjusts behavior to reduce overuse

Hardware Components

- ESP32 Development Board
- MAX9814 Microphone Module
- Piezo Vibration Sensor
- DHT22 Temperature/Humidity Sensor
- MQ135 Air Quality Sensor
- RGB LED Module
- Coin Vibration Motor
- Battery & Charging Circuit

